

Technology News

First human-like robot ready for space trip

The National Aeronautics and Space Administration (NASA) and General Motors, USA, plan to send the first human-like robot to the International Space Station to help astronauts. The robot, called Robonaut 2 or R2, weighs 140 kg and has a head, torso and two

Control electronic devices by thought

What if gadgets around us could read our thoughts and act accordingly? If reports are to be believed, something similar may happen very soon. A consortium of Japanese companies, research institutes and the government is working on these new generation

equipment smaller before it can be made commercially available.

Flying cars just a few years away

In a few years, flying cars might become reality. The Defence Advanced Projects Agency (DARPA) of USA has cleared the Transformer (TX) programme that will monitor the development of flying vehicles for the US military. If things fall in place, by the year 2015 US soldiers will have the ability to drive into battle in a flying car.

The Transformer will give troops the ability to avoid traditional and unconventional threats such as ambushes and improvised explosive devices and road obstructions. The Pentagon has earmarked \$54 million so far to develop the concept and DARPA is expected to take four years to develop a working prototype.

'Touch-floor' control for devices

Imagine your music system playing music of your choice as you slide your foot across a particular stretch of the floor of your room. While touchscreens for portable electronics are common, German scientists Patrick Baudisch and colleagues at the Hasso Plattner Institute have developed a touch-floor. The group's prototype, called 'Multi-toe,' is based on a technique known as frustrated total internal reflection (FTIR) that is already used in some touchscreens.

The Multi-toe is made up of a 0.5mm-thick sheet of silicone lying on an 8mm-thick layer of clear acrylic, both of which sit on a thick glass sheet to provide rigidity. Light beams shone into the acrylic bounce around inside until pressure from a foot, say, allows them to escape. A camera below captures the light and



Robonaut2, the next-generation dexterous robot, developed by NASA and General Motors

arms that function fully.

R2 will be sent onboard space shuttle Discovery, which is scheduled to head to the space station in September. NASA hopes to eventually use R2 to assist astronauts during space walks and to work alongside engineers inside the space station.

John Olson, director of NASA's Exploration Systems Integration Office, said, "The partnership of humans and robots will be critical to opening up the solar system and will allow us to go farther and achieve more than we can probably even imagine today."

electronic gadgets using advanced versions of the existing brain-machine interface (BMI) technology that will be able to read the user's mind and automatically respond to his wishes. So one could be able to change TV channels or send text messages by thought.

BMI machines have already been developed through which an operator can make a robot carry out an action, such as move an arm or lift a leg. The scientists also have the technology that lets the machine record the number thought of by a person.

The challenge now is to simplify the existing systems and make the